The `surfacePosition` function object: implementation and tests

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Abstract

`surfacePosition` is an OpenFOAM function object that reports the height and position of a volume-of-fluid interface at a set of locations, with the same entries and output as OpenFOAM's `interfaceHeight`, which integrates the interpolated volume fraction along a sampled line. `surfacePosition` finds the cells crossed by each line once and follows them through mesh changes by a flood fill, so its cost per write does not depend on the mesh size, and it sums the lengths of the line inside the liquid, cutting each interface cell geometrically with the isosurface of the volume fraction. This report describes the implementation and the test suite in `test`. All numbers, tables and figures are generated from the output of one run of the suite (wall time 1 min 44 s). In a 1 000 000-cell case on 18 processors, `surfacePosition` costs 4.5 ms per time step for 50 lines, against 1256 ms for `interfaceHeight`; on the exact stream function wave it is more accurate than `interfaceHeight` on the same mesh, and it is exact for flat interfaces, also on faces and processor boundaries.

1 Usage

`surfacePosition` is built into the library `libsurfacePosition` by `Allwmake` in its repository. It links OpenFOAM's `finiteVolume`, `meshTools`, `sampling`, `lagrangian` and `geometricVoF` libraries. Its entries and output are those of `interfaceHeight`, so that a dictionary of `interfaceHeight` works after changing its `type` and `libs`, and post-processing tools for `interfaceHeight` read its output:

```
surfacePosition1
{
    type            surfacePosition;
    libs            (surfacePosition);
    locations       ((0 0 0) (10 0 0) (20 0 0));
    alpha           alpha.water;    // default alpha
    liquid          true;           // alpha is the liquid's
    direction       (0 0 -1);       // downwards, default gravity
    mode            accumulated;    // accumulated, highest or lowest
    surfCellTol     1e-8;           // interface cells
    isoFaceTol      1e-8;           // tolerance of the isosurface cut
    checkLineCells  false;          // tests: compare line cells with all cells
    fullSearchInterval 0;           // full search every N mesh changes
}
```

The `interfaceHeight` entry `interpolationScheme` is not used. The files `height.dat` and `position.dat` in `postProcessing/<name>/<time>` hold, per location, the height h_B of the liquid (in the mode, below) above the lower boundary, where the line leaves the mesh along the direction, the height $h_L = h_B - h_{LB}$ of the interface above the location, where h_{LB} is the height of the location above the lower boundary, and the position of the interface, the location moved by h_L against the direction. As in `interfaceHeight`, a dry line gives $h_B = 0$, and a line missing the mesh $h_B = 0$ and $h_L =$ GREAT (10^{15}). The `mode` sets which height is reported: `accumulated`

(default), the length of the line in the liquid as in `interfaceHeight`; `highest`, the top of the highest liquid along the line, e.g. including droplets above the surface; or `lowest`, the top of the lowest liquid part, the first transition from liquid to gas going up the line. The headers of both files state the mode in a comment line (`# Mode :  accumulated`), so the columns are unchanged.

2 Implementation

2.1 Lines

Through each location a line runs along the unit downward direction $\mathbf{t}$ (the `direction` entry, or gravity), from above the mesh to below it, across the bounding box of the mesh plus a margin of the box diagonal for mesh motion, wherever the location is.

2.2 Cells crossed by a line

Whether a line $\mathbf{x}(s) = \mathbf{x}_0 + s\,\mathbf{t}$, $0 \le s \le L$, crosses a convex cell is decided by clipping the parameter interval against the half-spaces of the cell faces. For a face with outward area vector $\mathbf{S}_f$ and centre $\mathbf{x}_f$, let $a = \mathbf{S}_f \cdot \mathbf{t}$ and $b = \mathbf{S}_f \cdot (\mathbf{x}_f - \mathbf{x}_0)$. For $a > 0$ the interval is cut above at $s = b/a$ (the line leaves), for $a < 0$ below (the line enters). The face is taken as parallel to the line if the distance of the line from the face plane changes by less than a tolerance along the whole line,

$$|a|\,L \le \tau, \qquad \tau = 10^{-9}\,|\mathbf{S}_f|^{3/2}, \tag{1}$$

and the cell is then rejected only if $b < -\tau$. A line lying in a face plane is thereby inside the cells on both sides. The cell is crossed if the clipped interval has a length above $10^{-15}L$; its ends s_{in} and s_{out}, the segment of the line in the cell, are stored with the cells.

The cells are found once by a `faceOnlySet` with a cached `meshSearch`, plus the cells nearest to five points of the line inside the local mesh bounds, which catch lines lying on faces. These cells seed the flood fill below, which completes the set.

Flood fill. After a mesh change the cells are found again without a search tree, by a flood fill that tests the face neighbours of cells crossing the line. The seeds are the old line cells, mapped through `reverseCellMap` after a topology change (`updateMesh`) and unchanged after mesh motion (`movePoints`). If no seed crosses the line any more, rings of face neighbours around the seeds are tested, up to 10 rings, until a ring holds a cell crossing the line.

The flood fill and the ring search continue across processor faces: a candidate cell reaching a processor face sends the cell on the other side to the neighbouring processor, with the number of rings still to search around it. All lines are processed together, with one exchange per round, until no processor has candidates to send. A line that moves into a neighbouring processor subdomain is thereby followed (section 3.3). A line that moves further than the ring limit, or that is connected to its old cells only through cyclic faces, is only found by a full search, done every `fullSearchInterval` mesh changes if requested.

With `checkLineCells`, the cells found after every mesh change are compared with all cells of the mesh that cross the line, and the numbers of missing and extra cells are written to the log. The test suite uses this to verify the flood fill.

2.3 Liquid along a line

At each write, each cell of a line contributes the part of its segment $[s_{\text{in}}, s_{\text{out}}]$ that lies in the liquid:

- a full cell, $\alpha \ge 1 - \epsilon$ with $\epsilon =$ `surfCellTol`, its whole segment, without any computation;
- an empty cell, $\alpha \le \epsilon$, nothing;

- an interface cell, $\epsilon < \alpha < 1 - \epsilon$, the part of the segment on the liquid side of the isosurface of α, interpolated to the mesh points, that cuts the volume fraction α from the cell. The function object holds its own point values of α, interpolated once per write, and its own `surfaceIteratorIso`, whose `vofCutCell` finds the isosurface as in the `isoAlpha` reconstruction (tolerance `isoFaceTol`, 100 iterations). With the centre $\mathbf{c}$ and area vector $\mathbf{n}$ of the isosurface, oriented towards the centre of the submerged part of the cell, the liquid is where $(\mathbf{x}(s) - \mathbf{c}) \cdot \mathbf{n} > 0$; the segment is cut at $s = \mathbf{n} \cdot (\mathbf{c} - \mathbf{x}_0)/(\mathbf{n} \cdot \mathbf{t})$. A cell that `vofCutCell` does not cut is counted as full or empty by $\alpha \geq 0.5$.

The liquid parts of the cells on each processor are merged where they touch, so that a column of full cells becomes a single part, and gathered on the master processor.

2.4 Heights and positions

Where several cells cover the same part of a line (a line in a face plane, on an edge, or on a processor or cyclic face), their liquid lengths are averaged: after every mesh change the ends of the segments of all line cells, on all processors, divide the line into pieces, each covered by a known number of cells. The liquid length of the line is

$$W = \sum_{\text{liquid parts } p} \sum_{\text{pieces } j} \frac{|p \cap j|}{n_j}, \tag{2}$$

where n_j is the number of cells covering piece j, and the length of the line in the mesh is the total length of the covered pieces. As in `interfaceHeight`, the height of the liquid above the lower boundary is $h_B = W$ (or the length in the mesh minus W with `liquid false`), the lower boundary being the end of the last covered piece. A sharp interface on a face, as set by `setFields`, needs no special treatment, and droplets or bubbles shift the height only by their volume along the line.

For the modes `highest` and `lowest`, the line is divided at the ends of the pieces and of the liquid parts, and an interval is in the liquid if at least half of the cells covering it are. The top of the first liquid interval from above gives the highest point, and the top of the last run of liquid intervals the top of the lowest liquid part.

3 Test suite

The suite (`test/Allrun`) runs four tests, each with a summary log and a `results.json` read by this report:

Test	What it checks	Processors
`waveAccuracy`	accuracy against an exact wave	1 and 4
`specialCases`	interfaces and lines on faces, edges, processor and cyclic patches	1 and 8
`movingMesh`	line cells after adaptive refinement and mesh motion	1 and 4
`largeTiming`	cost in a large case	18

`largeTiming` runs on the last 18 cores, bound one process per core with local memory through an Open MPI rank file, while the other tests run on the remaining cores. The suite passes if the checks of `specialCases` and `movingMesh` pass; in the run reported here they passed and passed, respectively.

About `interfaceHeight`. `interfaceHeight` was added to OpenFOAM by the OpenFOAM Foundation in May 2017 (OpenFOAM-6) and has been in OpenFOAM.com since v2006, which added the optional `direction` entry. Its algorithm is the same in all OpenFOAM.com releases from v2006 to v2606: for each location it builds a `midPointAndFaceSet` along the mesh diagonal in

the direction, which points downwards (gravity), with a new `meshSearch` at every write, integrates the interpolated volume fraction along it and sums the result over the processors. The only changes are the writing of the file headers (removed in v2012, restored on the master processor in v2406) and the documentation (v2512). The OpenFOAM Foundation corrected its parallel operation in May 2023 (commit `1d78434bee`, OpenFOAM-11) by skipping sampling segments between two faces; OpenFOAM.com does not have this correction. The tests use the v2512 version.

3.1 Accuracy: exact stream function wave

`streamFunFoam` sets the fields to an exact stream function wave (height 0.125 m, depth 0.4 m, period 2 s) in a periodic flume one wave length long, at every time step, and a coded function object writes the exact surface elevation at the lines. The flume has a coarse (38×8) and a medium (95×20) mesh. Twelve vertical lines are placed in four cell columns, at the cell centre, a quarter cell off it and 2 % of a cell from a face (fig. 1). The elevations are compared over one period with 40 samples, for `surfacePosition` and for `interfaceHeight` with its default `cellPoint` and with `cell` interpolation.

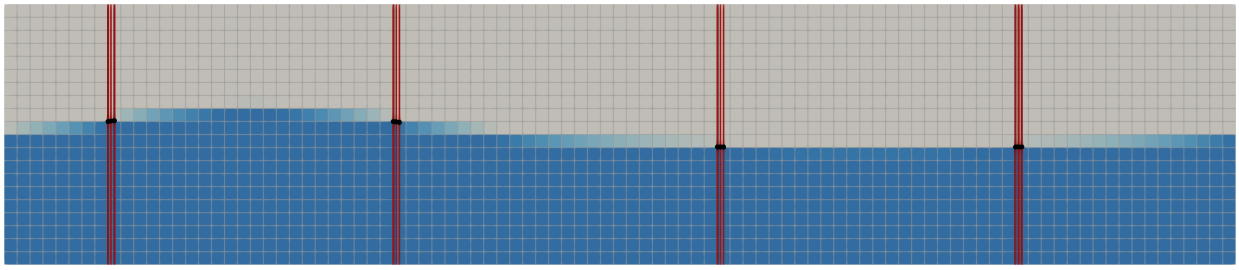


Figure 1: `waveAccuracy`, medium mesh at $t = 2$ s: volume fraction (white air, blue water), the 12 vertical lines (red, three per cell column) and the positions found by `surfacePosition`.

Table 1: Error of the interface elevation against the exact wave [mm]: rms per position of the line in the cell (in cell widths from the face), and mean and maximum magnitude over all lines and samples.

	rms, line position			all lines	
Method	0.50	0.25	0.02	mean	max
coarse, 38×8 cells					
`interfaceHeight`, cellPoint	0.499	0.562	0.511	0.000	1.631
`interfaceHeight`, cell	25.001	25.082	25.297	-25.000	31.670
`surfacePosition`	0.253	0.196	0.091	0.000	0.741
medium, 95×20 cells					
`interfaceHeight`, cellPoint	0.081	0.091	0.083	-0.000	0.267
`interfaceHeight`, cell	10.000	10.033	10.119	-10.000	12.679
`surfacePosition`	0.041	0.031	0.007	0.000	0.119

Table 1 shows that `surfacePosition` has the smallest error on both meshes: 0.192 mm rms on the coarse mesh against 0.525 mm for `interfaceHeight` (cellPoint), and 0.030 mm against 0.085 mm on the medium mesh, where its largest error is 0.119 mm. `interfaceHeight` with cell interpolation is biased by a quarter of the cell height (−25.0 mm and −10.0 mm); the sign of this bias follows the direction. Fig. 2 shows the errors over the period at a cell centre and 2 % of a cell from a face. The serial and parallel results are identical.

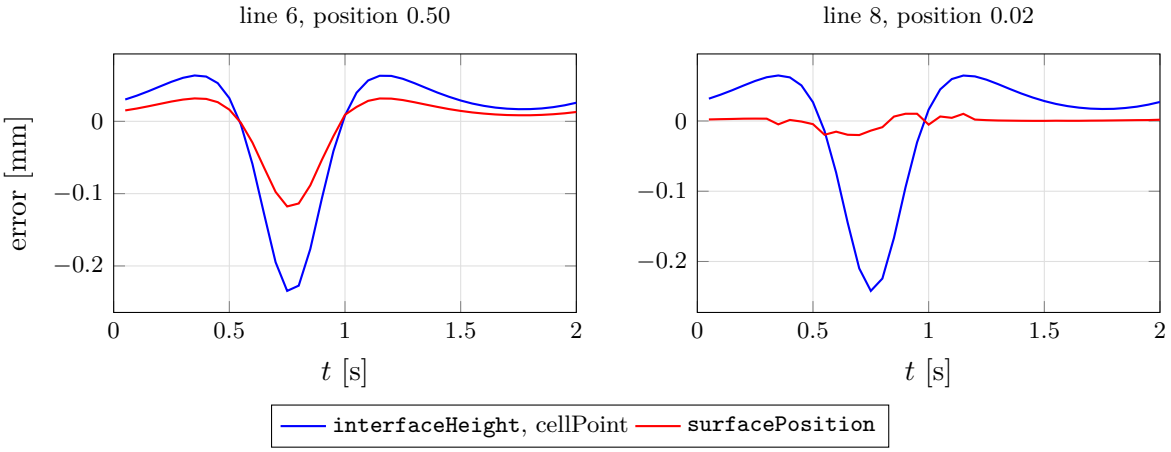
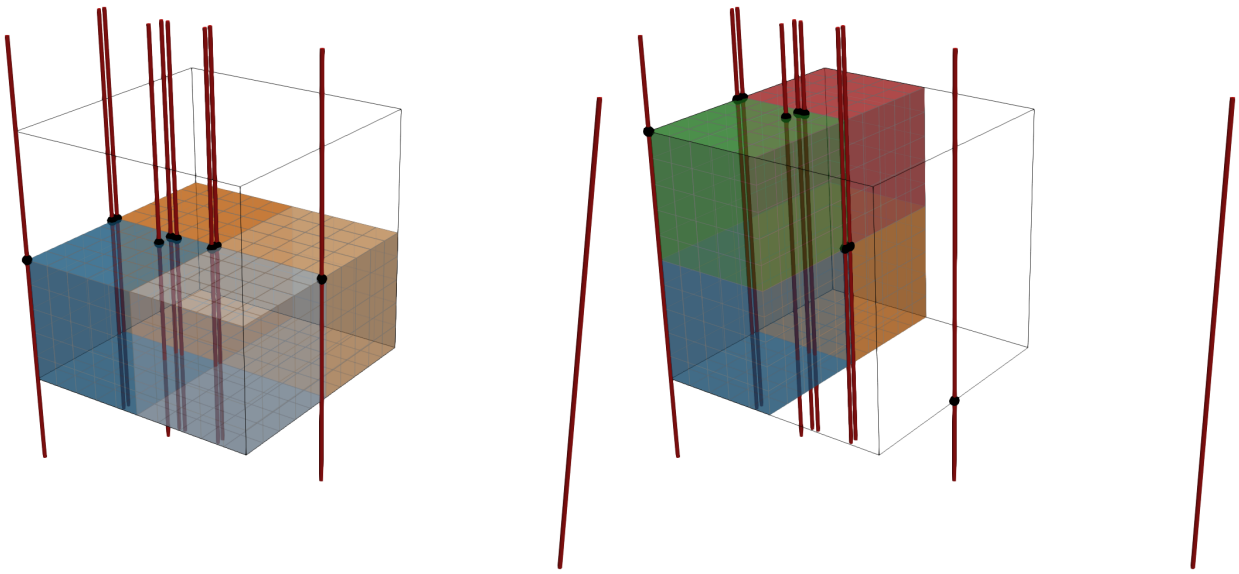


Figure 2: Error of the elevation over one wave period on the medium mesh, at a cell centre and 2 % of a cell width from a face.

3.2 Special cases

A closed $10 \times 10 \times 10$ cube, cyclic in x, decomposed into 8 subdomains, holds a flat interface ($z = 0.537$ inside cells, $z = 0.5$ on internal and processor faces, $z = 0.3$ on internal faces), a vertical slab of water ($x < 0.5$), whose interfaces lie on the cyclic and the processor faces, or two layers of water, below $z = 0.35$ and at $0.6 < z < 0.75$ with a sharp lower interface. After one short `interIsoFoam` step, twelve vertical lines (fig. 3) test locations in a cell, on a face plane, on an edge, on a processor face and edge, on the cyclic patches, on a cyclic/processor edge and a cyclic/wall edge, far above and far below the mesh, and outside the domain, with `surfacePosition` in its three modes. The heights and positions are compared with the exact values: for the flat interfaces $h_B = z$ in all modes on all lines crossing the mesh; for the slab $h_B = 1$ for $x < 0.5$, 0 for $x > 0.5$, and on the processor faces at $x = 0.5$, where a full and an empty cell meet, 0.5 accumulated and 1 in the other modes; for the layers $h_B = 0.5$ accumulated, 0.75 highest and 0.35 lowest; and the marker of a line missing the mesh for the line outside the domain.



(a) Flat interface at $z = 0.5$ (b) Vertical slab of water

Figure 3: `specialCases` on 8 processors: water coloured by processor, the vertical lines through the locations (red; the line on the right is outside the domain) and the positions found by `surfacePosition`.

Table 2: Largest error of the heights h_B, h_L and positions of `surfacePosition` in its three modes against the exact values [m], over the lines crossing the mesh (tolerance 1×10^{-8} m), and the number of lines correctly reported as missing the mesh.

Scenario	Proc.	accumulated	highest	lowest	off mesh
flat, $z = 0.537$	1	1×10^{-15}	1×10^{-15}	1×10^{-15}	1
	8	1×10^{-15}	1×10^{-15}	1×10^{-15}	1
flat, $z = 0.5$	1	0	0	0	1
	8	0	0	0	1
flat, $z = 0.3$	1	2.8×10^{-15}	2.8×10^{-15}	2.8×10^{-15}	1
	8	2.8×10^{-15}	2.8×10^{-15}	2.8×10^{-15}	1
vertical slab	1	2.2×10^{-16}	2.2×10^{-16}	2.2×10^{-16}	1
	8	2.2×10^{-16}	2.2×10^{-16}	2.2×10^{-16}	1
two layers	1	0	0	1.4×10^{-15}	1
	8	2.2×10^{-16}	2.2×10^{-16}	1.4×10^{-15}	1

All heights and positions are exact to round-off in all modes (table 2), in serial and on 8 processors, including the interfaces on processor and cyclic faces and the lines on faces, edges and patches. Table 3 shows h_B of `interfaceHeight` on some of the lines. On 8 processors, the lines on processor faces and edges count the water of the cells on all sides; on the cyclic patches and their edges it gives 0.5 instead of 1 for the slab; and for locations far above or below the mesh it gives 0, as its line, of the length of the mesh diagonal around the location, does not reach the mesh.

Table 3: `interfaceHeight`: height h_B [m] above the bottom on some of the special lines, for the flat interface at $z = 0.5$ and the slab.

		flat, $z = 0.5$			slab		
Line	Location	exact	serial	8 proc.	exact	serial	8 proc.
3	processor face	0.500	0.500	1.000	0.500	0.500	1.000
4	processor edge	0.500	0.500	1.975	0.500	0.500	1.750
5	cyclic patch	0.500	0.500	0.500	1.000	0.500	0.500
7	cyclic/processor edge	0.500	0.500	0.525	1.000	0.500	0.750
8	far above	0.500	0.000	0.000	1.000	0.000	0.000
9	far below	0.500	0.000	0.000	1.000	0.000	0.000
10	outside the domain	–	0.000	0.000	–	0.000	0.000

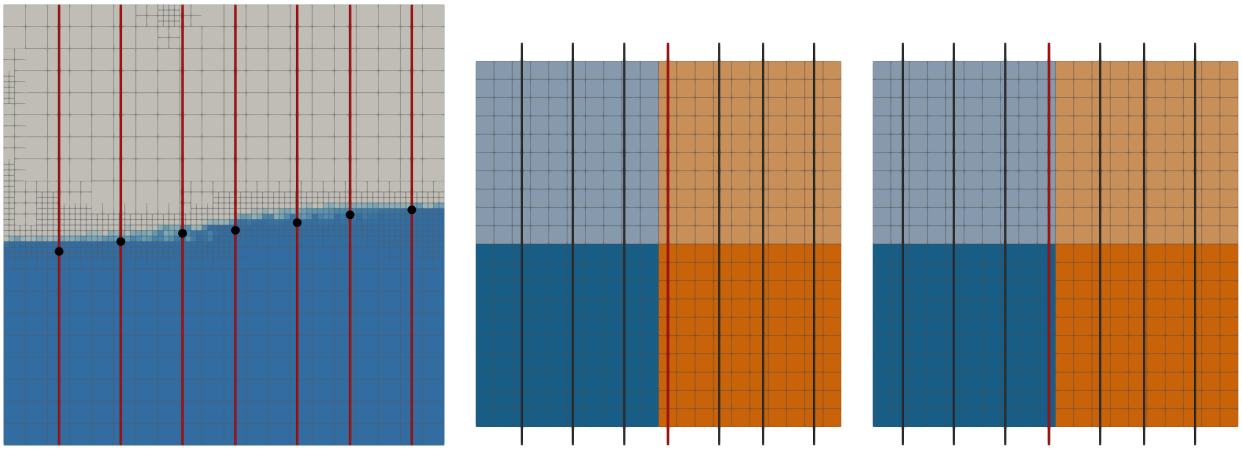
3.3 Moving meshes

A 2D standing wave (amplitude 0.05 m) in a closed unit box with a 20×20 mesh, one cell thick in y, is run with `interIsoFoam` in two variants, in serial and on 4 processors:

amr adaptive refinement of the interface cells to two levels at every time step with `dynamicRefineFvMesh`, for 400 time steps. The refinement also splits the cells in y, so that the seven lines, which lie in the plane $y = 0$, run in the internal faces of the refined cells (fig. 4a).

motion a solid body oscillation of the whole mesh in x, 0.05 m $\sin(5.3\,t)$, for 0.6 s, with the lines fixed in space, so that the line cells change at every time step.

In all runs the flood fill finds exactly the cells crossing the lines (table 4): 177 and 396 mesh changes in the serial and parallel amr runs, 50 in the motion runs, and no line misses the mesh. In the parallel motion run, line 3 moves from the subdomains right of a processor boundary to those



(a) amr, $t = 0.0818\,\mathrm{s}$ (b) motion, $t = 0$ (c) motion, $t = 0.2\,\mathrm{s}$

Figure 4: `movingMesh`. (a) Refined mesh, volume fraction, lines and the positions found by `surfacePosition`. (b, c) Mesh coloured by processor on 4 processors with line 3 (red): the mesh moves right, and line 3 passes from the right to the left subdomains.

Table 4: `movingMesh`: line cells after all mesh changes compared with all cells (`checkLineCells`), outputs of lines missing the mesh (off mesh), and the rms and largest difference of h_B from `interfaceHeight` [mm].

				line cells				vs. `interfaceHeight`	
Case	Proc.	Steps	Changes	found	missing	extra	off mesh	rms	max
amr	1	400	177	48730	0	0	0	6.52	42.1
amr	4	400	396	109857	0	0	0	7.26	27.4
motion	1	51	50	7000	0	0	0	0.43	3.0
motion	4	51	50	7000	0	0	0	0.43	3.0

left of it between $t = 0.1$ and $0.1125\,\mathrm{s}$ (fig. 4); the flood fill follows it across the processor faces and finds all 7000 line cells, as in serial.

In the amr runs the refinement produces spurious velocities that limit the time step to about $0.000\,21\,\mathrm{s}$ and throw up droplets. The heights of `surfacePosition` and `interfaceHeight` still differ by at most 42 mm, since both measure the liquid volume along the line; serial and parallel amr runs diverge (`interfaceHeight` differs by up to 10.8 mm). In the motion runs, serial and parallel flows differ slightly, by up to 0.13 mm in `interfaceHeight` and 0.21 mm in `surfacePosition`.

3.4 Cost in a large case

A 3D standing wave in a closed unit cube with 1 000 000 cells is run with `interIsoFoam` on 18 processors (scotch decomposition), with 50 vertical lines placed off the cell centres (fig. 5) and `surfacePosition` and `interfaceHeight` writing at every time step. OpenFOAM's profiling gives the time of each function object and of the flow solution; it is written with the fields after 5 start-up steps and at the end, and the difference gives the time of 10 steps.

`surfacePosition` costs 4.5 ms per step, 0.6 % of the flow solution, including the interpolation of α to the mesh points, while `interfaceHeight` costs 1256 ms, 170 % of the flow solution and 280 times as much as `surfacePosition` (table 5). The initial line search of `surfacePosition` takes about 1132 ms once. The heights of `surfacePosition` and `interfaceHeight` agree to 0.123 mm (rms 0.026 mm).

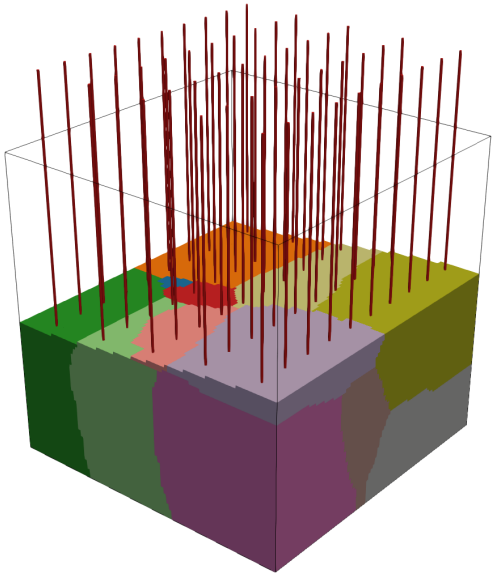


Figure 5: `largeTiming`: water region coloured by the 18 processor subdomains, and the 50 lines.

Table 5: Time per step (mean over the processors, each including the waiting for the others), per line, the range over the processors, and the start-up time of the first steps beyond the steady cost (the line search), in ms.

	per step	per line	range	start-up
flow solution (`interIsoFoam`)	739.4		1.41	
`surfacePosition`	4.5	0.090	1.77	1132
`interfaceHeight`	1256.2	25.124	0.70	

4 Findings of the tests

Two defects in the line cells of `surfacePosition` were found by the moving mesh tests and fixed:

1. *Round-off in the test for faces parallel to the line* (`crossesLine`). A face was taken as parallel to the line only if $|a| \leq 10^{-300}|\mathbf{S}_f|$. For the lines in the face planes $y = 0$ of the refined cells, round-off gave $|a|/|\mathbf{S}_f| \approx 10^{-17}$, so b/a became an arbitrary distance that could clip a crossed cell away and disconnect the flood fill. In a separate run of the serial amr case with the old test, all of its 177 mesh changes missed line cells; with eq. (1) none does.

2. *Lines lost across processor boundaries.* The flood fill followed internal faces only. When the motion moved line 3 into the neighbouring subdomains, neither side found any cell, and the line was lost for the rest of the run. The flood fill now continues across processor faces.

The checks of the suite were tightened accordingly: lines missing the mesh fail the moving mesh test, and the check scripts' exit codes are passed on to `Allrun`.

Earlier versions of `surfacePosition` reported the crossing of the line with the interface plane of one interface cell, the one with α closest to 0.5. In the amr runs, droplets with α near 0.5 and steep planes then gave positions up to metres outside the domain; the liquid length of section 2.4 removes this. The area vector of the isosurface from `surfaceIteratorIso`, documented to point from high to low values, gave the liquid on the wrong side in the flat test, so the liquid side is taken from the centre of the submerged part of the cell instead.

The tests also show that `interfaceHeight` (v2512) is wrong on processor faces and edges, on cyclic patches, and for locations outside the vertical extent of the mesh (table 3). The first may be what the OpenFOAM Foundation's parallel correction addresses (section 3); this was not tested.

5 Reproducing this report

With OpenFOAM v2512 sourced, in the repository of `surfacePosition`:

```
./Allwmake            # libsurfacePosition
cd test
./Allrun              # the test suite, results.json per test
cd report
./Allrun              # makeData.py, makeFigures.py (pvbatch), latexmk
```

`makeData.py` writes the numbers and tables in `generated/` from the `results.json` files and the `postProcessing` data of the cases; `makeFigures.py` renders the figures from the case data with ParaView. This report was generated from the run of 2026-09-17 11:49 (source and tests at revision `63c9f6f`).